# INORGANIC CHEMISTRY CUMULATIVE EXAM

## December 7, 1996

Give a rationale for the following experimental observations. Indicate what theory or model is the basis for your answer.

1) The electronic spectra of transition metal complexes frequently consist of classes of absorption bands with absorptivities ($a$ in units of L/mol-cm) values in the range $10^2$(visible) , $10^3$ (near UV) , and $10^4$ (UV). What is the probable origin of each class?

2) The visible absorption bands for Mn(II) complexes have $a \sim 0.01$ and Mn(III) complexes have $a \sim 10^2$ in the visible region.

3) Although most Co(III) octahedral complexes are diamagnetic, the anion $CoF_6^{3-}$ is blue and paramagnetic.

4) The visible spectra of many $d^1$ Ti(III) complexes show two partially resolved absorption bands in the visible region.

5) The anions $NiCl_4^{2-}$ and $Ni(CN)_4^{2-}$ are paramagnetic and diamagnetic respectively.

6) The visible spectrum of $Co(en)_3^{3+}$ has bands at 475 and 340 nm and the complex cis - $Co(en)_2 F_2^{1+}$ has bands at 560(shoulder) , 500 , and 370 nm. Qualitatively account for the shift in maxima and change in the number of bands.
[All bands have $a \sim$ 40 -80 L/mol-cm]